

Single-board Computers For Embedded Applications

Developments in VLSI technology have led to implementation of an entire microcomputer on a single board. Single-board computers (SBCs) today include latest microprocessors based on ARM and standard connections like LAN, PCIe, USB and Ethernet with minimal component count and smallest package size

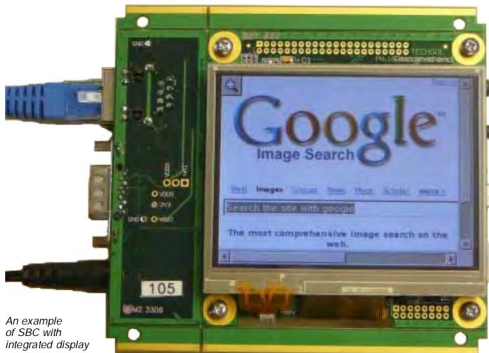
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One of the quickest and easiest ways to start building your project for demanding applications is to use single-board computers (SBCs). Unlike the motherboard of a personal computer that functions with various daughter boards to perform different tasks, an SBC combines all the features onto a single board that is as small as a Parle-G biscuit. Introduced in early 80s, SBCs have gained huge market acceptance for embedded applications like process control, industrial computing, data acquisition, research and development, and military projects.

"Although today's industrial and business markets are highly fragmented, there seems to be the need for an aggregated solution, i.e., embedded solution. Single-board computers help in producing a configurable, customised solution to address the targeted market," says Royson Rajan, vice president, Solusoft Technologies. SBCs are at the core of embedded solutions and have gained huge market as these come as a working board.

Some of the PC-compatible SBCs are based on the form-factor of IBM's ISA PC bus, while non-PC-based popular form factors include VME, PCI and PCIe. "Today's smallest SBCs are based on form factors like Mobile-ITX, Pico-ITX, Nano-ITX, Mini-ITX to EM-ITX boards," shares Sathya, director, Shresta eTechnologies.

ARM processors have also entered



An example of SBC with integrated display

the SBC space, which was earlier dominated by PowerPC and Intel architecture. Some of the latest SBCs also include multi-core processors for applications requiring intense computing.

SBC requirements

To serve constantly changing demands of the embedded market, time-to-market is a critical need. "Single-board computers help to address these demands by offering convenience, flexibility and scalability of commercial off-the-shelf products while at the same time incorporating latest chips and technology," says S. Manikandan, managing director of Pentagon Rugged Systems.

Size is an important requirement

for most SBC designers, as the computing device may not fit the location where it is going to be inserted. "The users normally have requirements regarding size (dimensions) of the board or appliance (box-level products)," says Rajan.

"The processor is the main component of the SBC. Hence it is essential to know the type of the processor, processing speed, etc," shares Rajesh V., managing director of MRR Embedded Systems. "The requirement for communication peripherals depends on the application area. However, designers are keen on knowing the availability of Ethernet, CAN bus, RS232, RS485, Wi-Fi, Bluetooth, etc. They also want information related to display options like